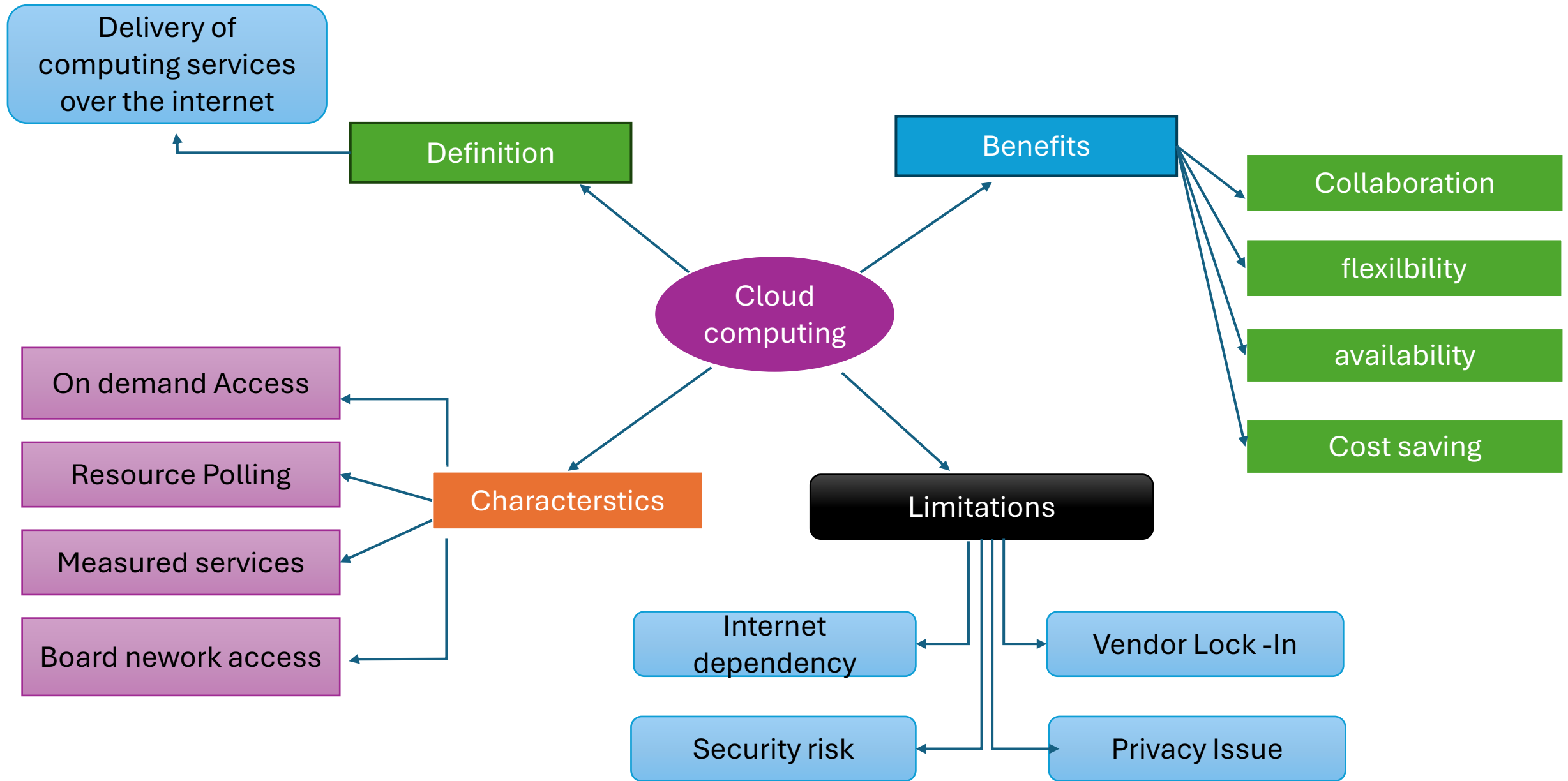




EVOLUTION OF CLOUD COMPUTING

Hardware Evolution

- Mainframes
- Client server
- virtualization

Internet Evolution

- Web application
- High speed internet
- Web services

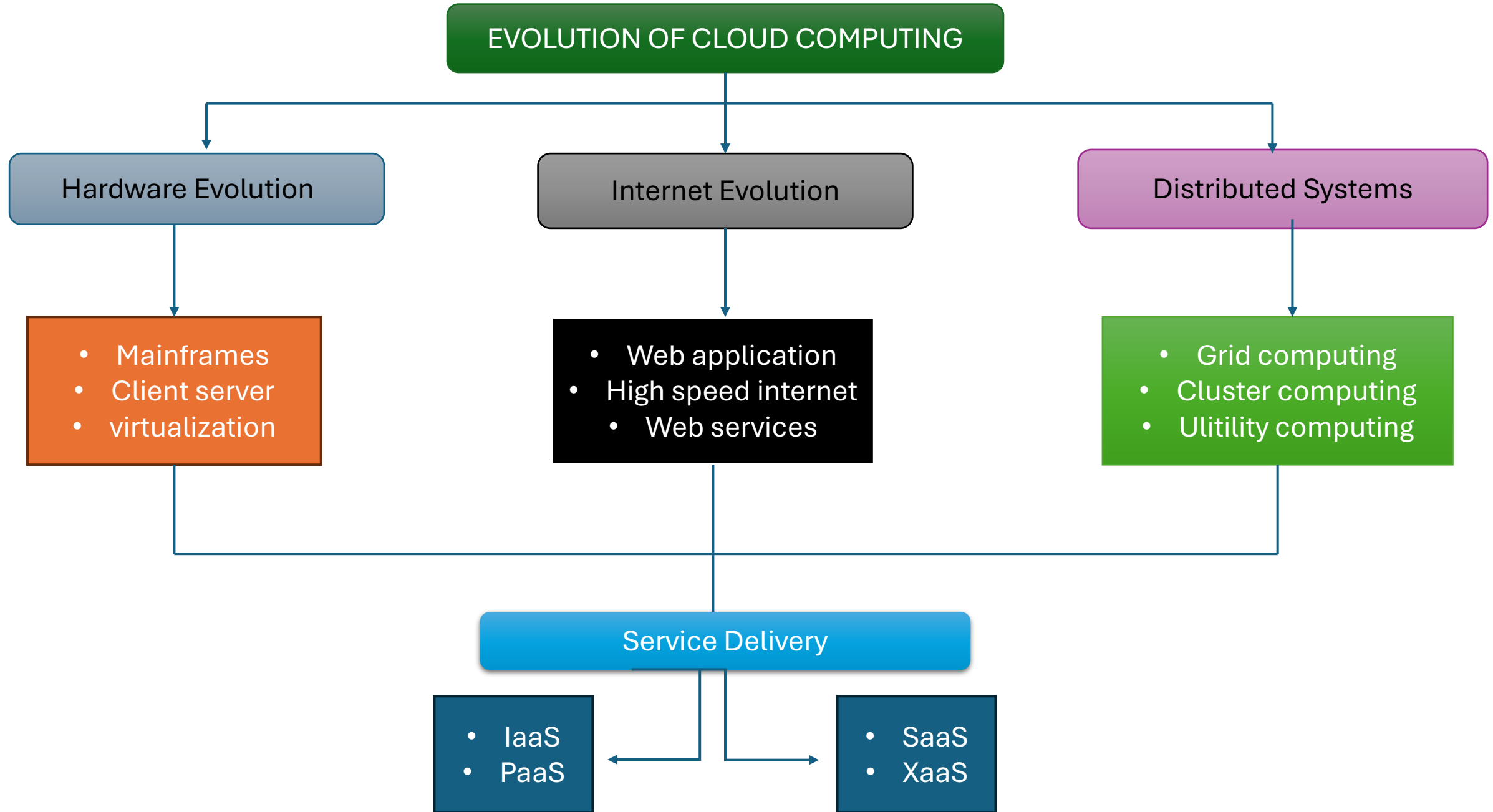
Distributed Systems

- Grid computing
- Cluster computing
- Utility computing

Service Delivery

- IaaS
- PaaS

- SaaS
- XaaS



CLOUD SERVICE MODELS

```
graph TD; A[CLOUD SERVICE MODELS] --> B[IaaS  
Infrastructure as a Service]; A --> C[PaaS  
Platform as a Service]; A --> D[SaaS  
Software as a Service]; A --> E[XaaS  
Everything as a Service]; B --> F[PROVIDER:  
Servers  
Storage  
Network]; C --> G[PROVIDER:  
Runtime  
Database  
OS]; D --> H[PROVIDER:  
Application  
Maintenance  
Updates]; E --> I[PROVIDER:  
All Services];
```

IaaS
Infrastructure as a
Service

PROVIDER:
Servers
Storage
Network

PaaS
Platform as a Service

PROVIDER:
Runtime
Database
OS

SaaS
Software as a Service

PROVIDER:
Application
Maintenance
Updates

XaaS
Everything as a Service

PROVIDER:
All Services

VIRTUALIZATION & WEB SERVICES

Virtualization

Web Services

Creates Virtual
Machines (VMs)
Resource Sharing
Server Consolidation

Communication
using APIs
SOAP / REST
Data Exchange

Enable Cloud Computing

Faster Deployment

Scalability

Resource Utilization

